

AOD Grade-Wise Addition Pattern

Process Mapping Document

Company: Jindal Stainless Ltd | Department: Steel Melting Shop, AOD Operations

Document Type	Process Mapping Document
Business Process	Grade-wise AOD addition pattern for stainless steel production
Process Families Covered	300 Series, 400 Series, and 200 Series
Primary Users	Shift-In-Charge, AOD Operator, Production Team, Quality Team
Purpose	To map the grade-wise workflow, control points, inputs, and outputs for AOD operations

Document Purpose

This document presents the AOD addition pattern. The process maps show the sequence of activities, key material inputs, control points, and end-stage checks for different stainless steel grade families.

The document supports process understanding, standardisation, stakeholder communication, shift-level training, and operational governance.

1. Process Mapping Scope

The process maps cover the grade-wise AOD flow from zero batch and pouring through main blow, flux addition, sampling, stirring, and reduction. Each process map shows the operational sequence and process control window for one grade family.

Process Map ID	Grade Family	Trigger	Key Inputs	Output
PM-01	300 Series	Heat ready for AOD processing	Flux, Dololime, HCFeCr, HCFeNi, DRI / LCFeNi, HCFeMn	Heat ready for reduction stage
PM-02	400 Series	Heat ready for AOD processing	Flux, Dololime, HCFeCr, DRI, HCFeMn	Heat ready for reduction stage
PM-03	200 Series	Heat ready for AOD processing	Flux, Dololime, HCFeCr, DRI, HCFeMn	Heat ready for reduction stage

2. Process Mapping Legend

Start / Trigger	Flux or Lime Activity	Alloy / DRI Activity	Control / Check
Process start point or entry activity	Flux, Lime, or Dololime addition stage	Metallic addition stage such as HCFeCr, HCFeMn, DRI, HCFeNi, or LCFeNi	Temperature, stirring, reduction, or process window control

PM-01: 300 Series AOD Addition Pattern Process Map

This process map shows the 300 series addition flow, including HCFeCr, HCFeNi, DRI / LCFeNi, HCFeMn, flux addition up to 12 MT, sampling, stirring, and reduction control.

AOD Addition Pattern - 300 Series

Process Mapping Document

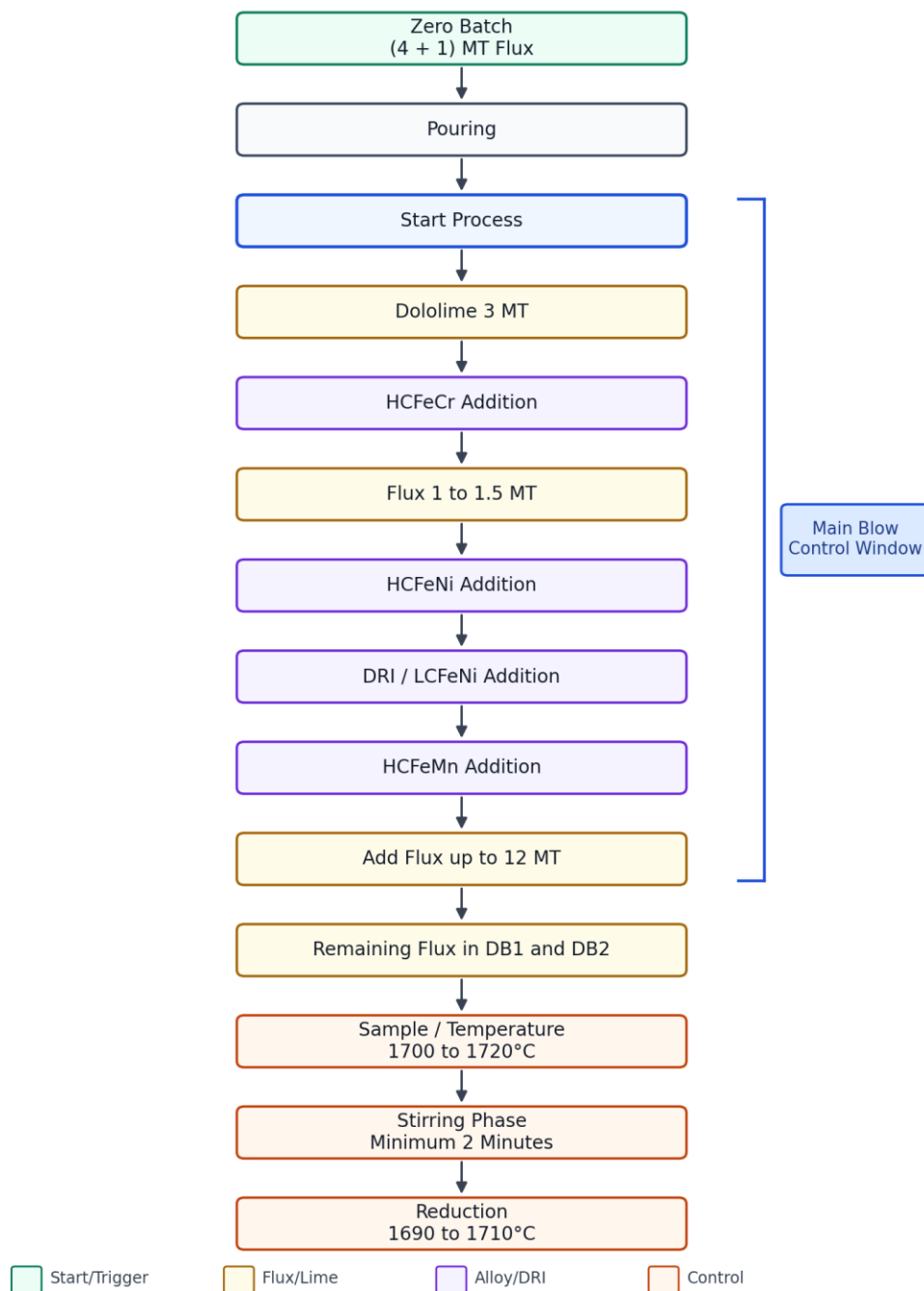


Figure PM-01: 300 Series AOD Addition Pattern Process Map

PM-02: 400 Series AOD Addition Pattern Process Map

This process map shows the 400 series addition flow, including HCFeCr, DRI, HCFeMn, flux addition up to 12 MT, sampling, stirring, and minimum reduction control.

AOD Addition Pattern - 400 Series

Process Mapping Document

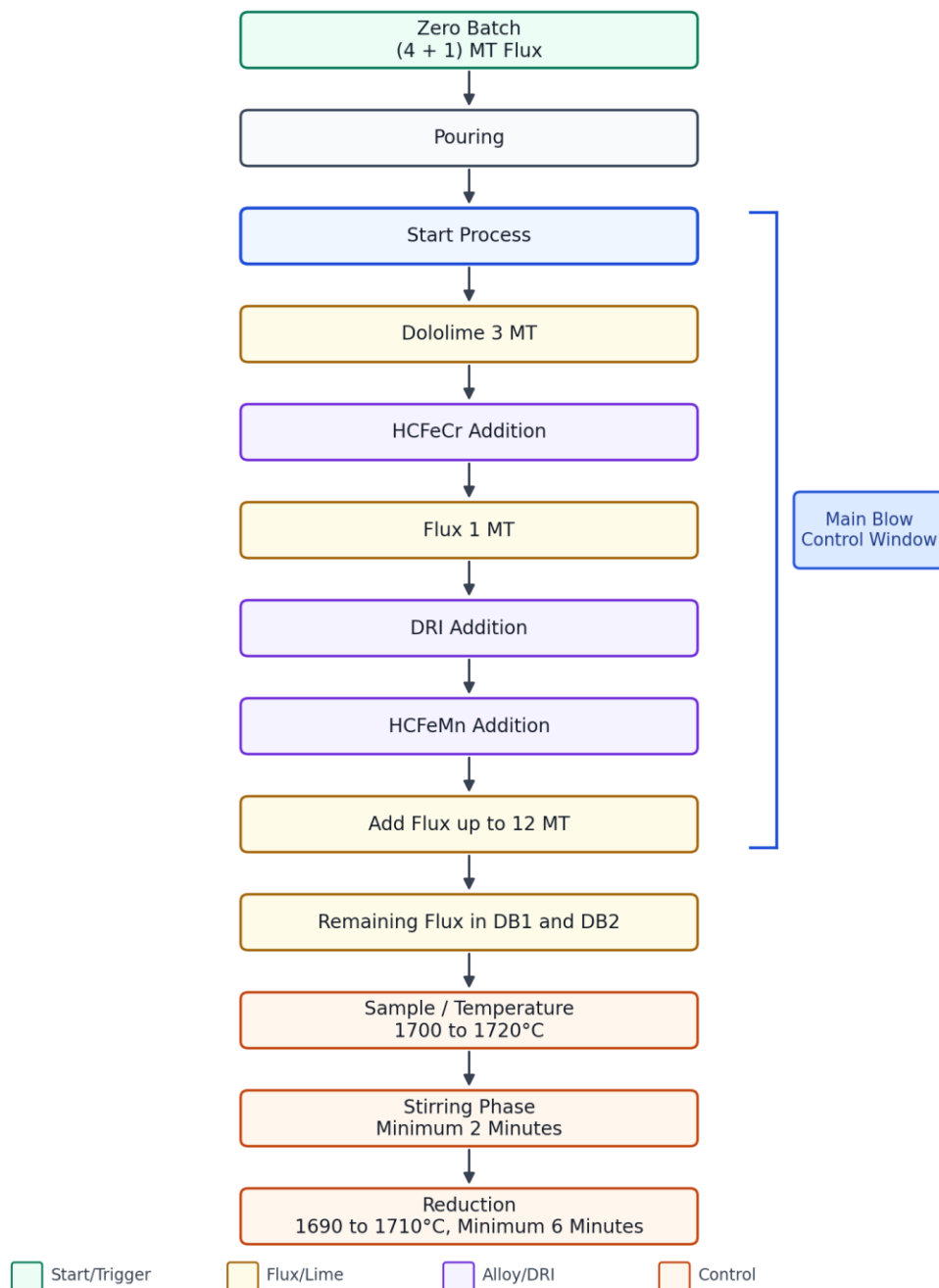


Figure PM-02: 400 Series AOD Addition Pattern Process Map

PM-03: 200 Series AOD Addition Pattern Process Map

This process map shows the 200 series addition flow, including HCFeCr, DRI, HCFeMn, higher flux addition up to 14 MT, sampling, stirring, and minimum reduction control.

AOD Addition Pattern - 200 Series

Process Mapping Document

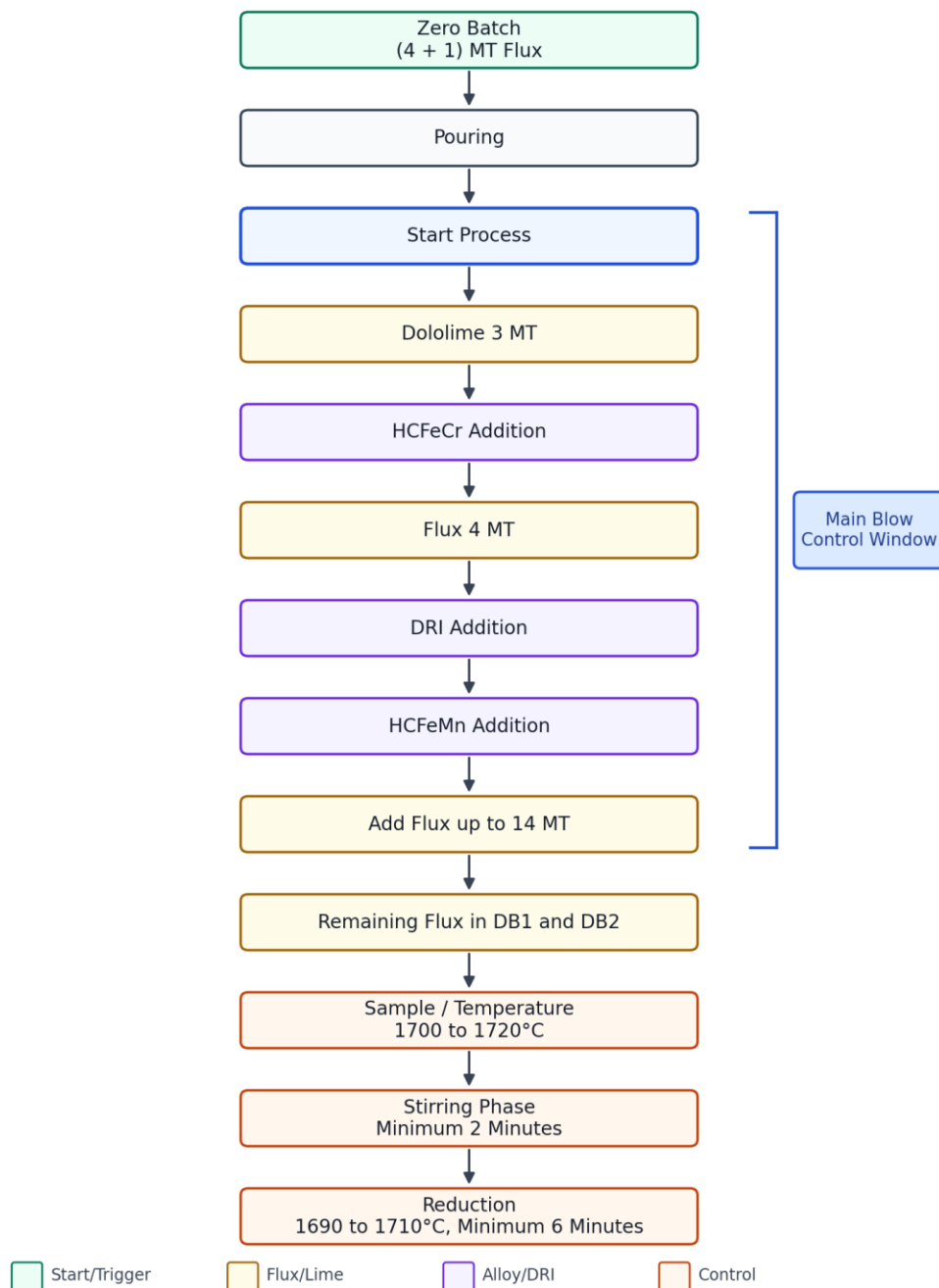


Figure PM-03: 200 Series AOD Addition Pattern Process Map

6. Business Rules Captured from Process Maps

Rule ID	Business Rule
BR-01	Each grade family must follow its own mapped AOD addition pattern.
BR-02	Zero batch and pouring must be completed before process start.
BR-03	Dolomite addition must occur after process start.
BR-04	Flux quantity must follow the applicable grade-family process map.
BR-05	Remaining flux must be added in DB1 and DB2 after the main blow flux limit is reached.
BR-06	Sampling temperature should be controlled in the defined operating range.
BR-07	Stirring phase must be completed before reduction.
BR-08	Reduction must follow the defined temperature and minimum time control where applicable.

7. BA Skills Demonstrated

BA Skill	How It Is Demonstrated
Process Mapping	Converted operational steps into visual process maps for each grade family.
Workflow Analysis	Identified the sequence of activities from zero batch to reduction stage.
Business Rules Documentation	Captured rules for flux addition, temperature control, stirring, and reduction.
Stakeholder Communication	Created a document that can be understood by production, quality, and shift teams.
As-Is / To-Be Thinking	Moved from experience-based execution to a structured process reference.
Operational Governance	Defined controls that support shift compliance and process consistency.